Chapter 1
Automatically Identifying Process Automation Candidates Using Natural Language Processing

Abstract The general goal of automation is to relieve humans from repetitive and routine-like tasks. The positive effects of automation have been demonstrated in various contexts and range from efficiency gains to the reduction of errors. In this chapter, we focus on the automation of individual tasks in a process using so-called software robots, which is often referred to as robotic process automation (RPA). More specifically, we focus on the task of identifying suitable candidates for such automation efforts. In practice, this identification task is associated with substantial manual effort and, hence, is both time and cost intensive. Recognizing these issues, we consider how also the identification of automation candidates itself can be supported through automation. We particularly focus on the way in which natural language processing (NLP) may be employed for this purpose. We show how NLP techniques support the identification of automation candidates in widely used process representations, such as process models and textual process descriptions. As such, we demonstrate how tackling one of the key impediments to the adoption of RPA may be supported in an algorithmic and automated manner.

1.1 Introduction

The goal of *automation* is often described as "*taking the robot out of the human*", i.e., relieve humans from repetitive and routine-like tasks. The positive effects of automation have been demonstrated in many contexts and range from efficiency gains to the reduction of errors [2]. Therefore, the interest in automating parts or even entire business processes is on the agenda of many organizations.

Taking a business process perspective, automation can be approached from two main angles: workflow automation and robotic process automation. *Workflow automation* is primarily concerned with automating the overall process coordination. This is typically achieved by using Business Process Management Systems (BPMSs) that enable, monitor, and manage the execution of business processes throughout an organization. By contrast, *Robotic process automation* (RPA) focuses

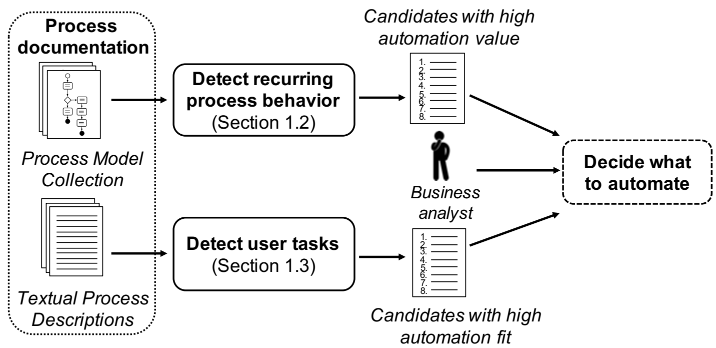

Fig. 1.1 Visualization of the manner in which the discussed approaches support business analysts in the identification of automation candidates

on the automation of individual tasks in a process using so-called software robots. Naturally, these two perspectives are complementary: a task that was automated by means of RPA might be easily integrated into a BPMS.

In this chapter, we focus on the second angle, i.e., on the automation of process tasks through RPA. A key question that arises in this context is: *Which process tasks should be automated?* In fact, this has turned out to be a major challenge in automation projects [21]. Intuitively, the suitability of a process task for automation endeavors can be considered from the perspectives of its automation fit and value. Here, *automation fit* is concerned with how easy it is to automate a particular task, while the *automation value* of a task is determined by considering how much time or money can be saved by automating it. A problem, though, is that assessing the automation potential of tasks may come with substantial manual effort and, hence, be both time and cost intensive.

To alleviate this problem, we use this chapter to demonstrate how also the identification of automation candidates itself can be supported through automation, thus reducing the required time and effort. Whereas various perspectives can be considered for this, we particularly focus on the way in which natural language processing (NLP) may be employed for this purpose. As visualized in Figure 1.1, we discuss two manners in which NLP techniques can support business analysts during the identification of automation candidates. Particularly, we consider (1) how natural language-driven analysis of process model collections can be used to detect recurring process behavior, which reveals candidates that have a high automation value, and (2) how textual process descriptions can be analyzed in order to detect so-called user tasks, which are recognized as candidates with a high automation fit.

In the remainder, Sections 1.2 and 1.3 discuss the general procedures of the two aforementioned approaches to support the identification of RPA candidates using NLP. Then, Section 1.4 positions this natural language-driven identification in light of other, complementary works that consider alternative process perspectives for the same purpose. Finally, Section 1.5 gives an outlook on what to expect from a technological perspective in the area of process automation in the coming years.

1.2 Identification based on Process Model Collections

This section shows how natural language-driven techniques can be leveraged to detect automation candidates in process model collections. The presented approach specifically addresses the perspective of *automation value* since it identifies (groups of) activities that commonly recur in an organization and, thus, would benefit more from automation.

1.2.1 Core Idea

Over the last years, many organizations have heavily invested in formally capturing their operations using business process models. As a result, these organizations maintain process model collections often containing hundreds or even thousands of process models. Given the insights that such collections can provide into the operations of an organization, they can serve as a valuable starting point for the identification of automation candidates. This is particularly the case because they highlight activities (or groups thereof) commonly occur within an organization's processes. Given that the potential automation value of an activity is higher when it occurs more frequently [23], such commonly occurring activities can represent promising automation candidates.

Recognizing this, we propose to employ our earlier work on the detection of frequently occurring activity patterns [18] as a means for the automatic identification of automation candidates. This natural language-driven approach achieves this in the manner described next.

1.2.2 Conceptual Approach

As illustrated in Figure 1.2, the approach comprises three main steps. Given a process model collection, the proposed approach first annotates the natural language labels contained in process models with information on the described actions and business objects. Second, the approach groups together activities that relate to equal or similar actions or business objects. Finally, the approach establishes a ranking of the most commonly occurring activity groups, which serves as a list of candidates that have a high potential automation value.

Natural Language Analysis of Process Models. Process models convey a major share of their meaning through the natural language labels associated with elements such as its activities. To identify automation candidates, our approach, therefore, targets the information contained in these labels. In this first step, we set out to annotate process model activity labels with information on the described *actions* and *business objects*. For instance, given a "*Deliver order*" activity, annotations sets

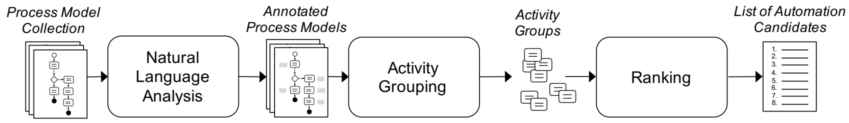

Fig. 1.2 Approach for the identification of automation candidates based on process model collections

out to identify the activity's action ("*deliver*") and the business object to which it is applied ("*order*").

This annotation task, however, is challenging because process model activity labels are short and barely ever represent proper sentences. As an example, consider the activity label "*Order verification*". From a grammatical point of view, both "*order*" and "*verification*" are nouns. From a semantic perspective, the noun "*verification*" implies the action "*to verify*" and the noun "*order*" represents a business object. For some activities, the appropriate annotation is less clear. As an extreme example, consider the activity label "*Plan data transfer*". Without further context, also a human is not able to derive whether the implied action is "*to plan*" or "*to transfer*". Since standard NLP techniques are not able to address this task, we developed dedicated parsers in prior work [15, 19]. They take process model activity labels as input and automatically recognize the described actions and business objects. This annotation, then, serves as input for the next step.

Activity Grouping. Building on the annotated process model labels, we propose different strategies to identify automation candidates:

- *Atomic identification:* This strategy employs the strictest notion to establish activity groups, since it only gathers activities if they share the same action and business object. Given that this assessment is based on the annotations established in the previous step, this strategy still allows for the grouping of activities that described the same activity in distinct manners. This strategy builds on the assumption that frequently occurring activities represent promising automation candidates.
- *Object-based identification:* This strategy abstracts from the identical action and focuses on activity groups that share same business object. As an example, consider the activities "*Check order*" and "*Approve order*". Since they share a common business object, these (and - if existing - other activities with the business object "*order*") would be grouped to a object-based automation candidate. The intuition behind this strategy is that the scope of a single automation candidate might not be limited to a single activity and, therefore, may include several related activities.
- *Hierarchy-based identification:* This strategy provides a further abstraction by grouping together activities that relate to business objects that are considered to be in a hierarchical relation, implied by compound nouns. As an example, consider the business objects "*order*" and "*purchase order*". Intuitively, one could argue that "*purchase order*" is a specific type of "*order*" and, therefore, activ-

ities with these hierarchically related business objects can be grouped together as well.

Finally, each of these strategies can also be transformed from a strategy that only groups activities when terms are fully equal into ones that incorporate semantic similarity considerations. This makes the strategies less strict, since they strive for the identification of similar, rather than equivalent actions or business objects. For example, the atomic-identification strategy would group "*Check order*" and "*Evaluate order*" activities together, because "*check*" and "*evaluate*" are actions that have a high semantic similarity, i.e., that have a similar meaning.

Ranking. The final step is the ranking of the identified activity group-based automation candidates. The intuition of the ranking step is that commonly occurring candidates are expected to be relevant than less common ones. Therefore, the automation candidates are ranked based on the number of activity instances they are based on. For candidates derived using the atomic identification strategy this is simply the number of occurrences of the considered activity. For candidates derived using the object or hierarchy-based identification strategy that is the number of occurrences of all activities that have been included in the respective group.

The approach's outcome is an ordered list of automation candidates. While this list still needs to be assessed by a domain expert, it provides an effective starting point for RPA efforts, since it reveals process steps whose automation would most likely result in larger gains across an organization.

1.3 Identification based on Textual Process Descriptions

This section shows how natural language-driven techniques can be leveraged to detect automation candidates in textual process descriptions. The presented approach specifically addresses the perspective of *automation fit* since it analyzes to what extent a particular task is actually automatable.

1.3.1 Core Idea

Next to process models, many organizations also maintain textual process descriptions. The reason is that textual descriptions can be created, read, and understood by virtually everyone since no specific knowledge about a modeling notation is required. Similar as for process model collections, textual process descriptions are a valuable starting point for the identification of automation candidates. Since they often provide a higher level of detail [16], this makes them excellent candidates for determining automation fit.

The core idea of the approach considered here is that it is possible to automatically identify so-called *user tasks* in these textual process descriptions. Such user tasks are defined as process activities in which a human interacts with an IT system, which makes them prime candidates for automation using RPA [5]. The approach presented next identifies these tasks through classification, distinguishing user tasks from manual tasks, i.e., activities that do not involve IT systems, and automated tasks, i.e., activities that do not require human involvement.

1.3.2 Conceptual Approach

Figure 1.3 illustrates the main steps of this approach. Below, we provide a brief technical explanation of each step. For a detailed description of the approach, we ask the interested reader to refer to [17].

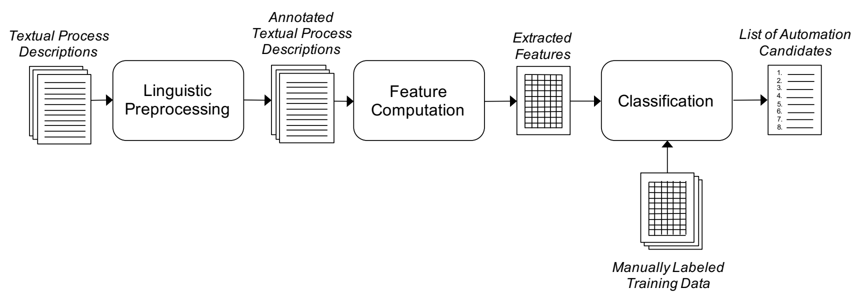

Fig. 1.3 Approach for the identification of automation candidates based on textual process descriptions

Linguistic Preprocessing. The goal of the linguistic preprocessing step is to automatically extract verbs, objects, and roles related to tasks described in the textual process description. To accomplish this, we build on a technique that was originally developed for the extraction of process models from natural language text [6]. This technique combines linguistic tools such as the Stanford Parser [11] and VerbNet [20] to, among others, identify verbs, objects, and roles. The advantage of this technique is its high accuracy and its ability to resolve so-called anaphoric references such as "*it*" and "*they*". To illustrate the output of this step, consider the sentence "*The vacations request process starts when an employee submits a vacation request via the ERP system*". From this sentence, we would automatically derive that the only relevant verb is "*submit*", that this verb relates to the object "*vacation request*", and that it is executed by the role "*employee*". This annotation serves as input for the next step.

Feature Computation. The selection and computation of suitable features is the key task when building a machine learning-based solution. Therefore, we manually analyzed a real-world data set to derive which characteristics determine the automation type of a task. As a result, we selected and implemented four features: 1) *verb feature* (categorical): What verb is used?, 2) *object feature* (categorical): What object is used?, 3) *resource type* (binary): Is the resource human or not?, 4) *IT domain* (binary): Does the task relate to the IT domain?

Classification. In the final step of our approach, the actual classification of tasks from process descriptions takes place. As described above, there is not a single feature that independently reveals the task type. It rather depends on the specific context of the task in the process. To be able to still classify unseen tasks, we employ a Support Vector Machine (SVM), a supervised machine learning algorithm. The advantages of SVMs are, among others, that they can deal well with relatively small datasets, they have a low risk of overfitting, and they scale well. Building on SVMs, our approach is able to automatically classify each task identified in the textual process descriptions with reasonable accuracy. As a result, we can automatically obtain a reliable overview of which activities are likely to be automatable.

1.4 Complementary Perspectives to Automation Identification

The previous sections considered the identification of automation candidates via natural language analysis of the labels and descriptions of processes and their activities. This perspective is complementary to existing works that consider the identification of automation candidates from other angles, such as approaches that aim to identify candidates based on recorded event sequences, represented in the form of *event logs*, which capture the execution of business processes [23], or as *interaction logs*, which capture the way in which users employ an information system to perform their tasks [4, 9]. These approaches cover a variety of aspects to identify suitable candidates in terms of automation value and fit.

Automation Value. To identify the (potential) automation value of activities, other works primarily consider how often a particular process activity is actually performed [23], arguing that RPA efforts should focus on frequently occurring activities. This view on the automation value of a process activity provides an interesting complement to the approach described in Section 1.2. This natural language-driven approach considers the commonality of activities across the processes that exist in an organization, answering the question: *in how many processes does this activity occur?* By contrast, when event logs are available, the analysis of execution frequencies, then, can answer the question: *how often does this activity actually occur within a particular process?* By combining these perspectives, one can thus identify activities that are both common in terms of their presence in various processes and in terms of their execution volume.

Automation Fit. The automation fit of activities is analyzed based on various aspects, these aspects cover process perspectives such as time (e.g., execution time [22]), data (e.g., how data is transformed in a step [14])), and resources (e.g., the number of involved actors [12]). However, the most commonly considered indicator of automation fit is the degree of process standardization [23]. Here, standardization can be considered in various ways, for instance in terms of the determinism of execution in terms of control-flow [4] and data transformations [14, 10], as well as from a process-output perspective, e.g., by looking at the failure rate [3], proneness to human error [8], and outcome predictability [22].

The classification approach described in Section 1.3 can be lifted to event logs and, thus, be used alongside the aforementioned approaches. For instance, approaches that determine the degree of standardization may be combined with the natural language-driven recognition of user tasks, allowing for the identification of activities that are both deterministic and involve the interaction of a user with an information system (as opposed to, e.g., fully manual activities). While this latter aspect could also be recognized by approaches that analyze resource involvement, these approaches can only be applied when an event log explicitly captures such characteristics, whereas natural language analysis can be used to overcome this requirement by deriving such information from the labels associated with an event.

1.5 Conclusion and Outlook

In this chapter, we demonstrated the opportunities of using NLP techniques for the identification of automation candidates in business processes. We showed that NLP techniques allow us to exploit widely used artifacts, in the form of process models and textual descriptions, to identify potential automation candidates. These candidate activities are intended as suggestions and, as such, serve as input for a final decision by experts on which of the candidates to actually automate using RPA.

On top of supporting the identification of automation candidates, NLP techniques are also valuable in other aspects of an RPA lifecycle. Particularly, they can also support both the design and execution of automation routines. The *design* of routines is typically concerned with the formalization of the to-be-automated task in the context of a flowchart. NLP techniques, particularly so-called text-to-model transformation techniques [1, 7], can provide valuable support to human designers in this regard. Such techniques are able to transform a textual description of a task into a formal process representation. While post-hoc analysis of the result is still required, these techniques are a helpful starting point for the formal definition of routines. In the context of the *execution* of automation routines, NLP techniques can help to push the boundaries of what can be automated. Here, NLP facilitates the automatic processing of both semi-structured documents, such as invoices, insurance claims, and contracts, as well as unstructured input, such as e-mails, phone calls, and chats. This move towards the automation of non-trivial tasks is widely referred to *cognitive automation* [13]. It can be expected that NLP and other techniques from the area of artificial intelligence will further contribute to the automation of work. While the boundaries are hard to predict, it is clear that we are getting closer to taking the robot out of the human.

References

1. Van der Aa, H., Di Ciccio, C., Leopold, H., Reijers, H.A.: Extracting declarative process models from natural language. In: International Conference on Advanced Information Systems Engineering, pp. 365–382. Springer (2019)
2. Aguirre, S., Rodriguez, A.: Automation of a business process using robotic process automation (rpa): A case study. In: Applied Computer Sciences in Engineering, pp. 65–71 (2017)
3. Beetz, R., Riedl, Y.: Robotic process automation: Developing a multi-criteria evaluation model for the selection of automatable business processes (2019)
4. Bosco, A., Augusto, A., Dumas, M., La Rosa, M., Fortino, G.: Discovering automatable routines from user interaction logs. In: International Conference on Business Process Management, pp. 144–162. Springer (2019)
5. Dumas, M., La Rosa, M., Mendling, J., Reijers, H.A., et al.: Fundamentals of business process management, vol. 1. Springer (2013)
6. Friedrich, F., Mendling, J., Puhlmann, F.: Process Model Generation from Natural Language Text. In: Proceedings of the 23rd international conference on Advanced Information Systems Engineering, *LNCS*, vol. 6741, pp. 482–496. Springer (2011)
7. Friedrich, F., Mendling, J., Puhlmann, F.: Process model generation from natural language text. In: International Conference on Advanced Information Systems Engineering, pp. 482–496. Springer (2011)
8. Ivančić, L., Vugec, D.S., Vukšić, V.B.: Robotic process automation: Systematic literature review. In: International Conference on Business Process Management, pp. 280–295. Springer (2019)
9. Jimenez-Ramirez, A., Reijers, H.A., Barba, I., Del Valle, C.: A method to improve the early stages of the robotic process automation lifecycle. In: International Conference on Advanced Information Systems Engineering, pp. 446–461. Springer (2019)
10. Jin, Z., Anderson, M.R., Cafarella, M., Jagadish, H.: Foofah: Transforming data by example. In: Proceedings of the 2017 ACM International Conference on Management of Data, pp. 683–698 (2017)
11. Klein, D., Manning, C.D.: Accurate Unlexicalized Parsing. 41st Meeting of the Association for Computational Linguistics pp. 423–430 (2003)
12. Kokina, J., Blanchette, S.: Early evidence of digital labor in accounting: Innovation with robotic process automation. International Journal of Accounting Information Systems **35**, 100,431 (2019)
13. Lacity, M., Willcocks, L.P.: Robotic process and cognitive automation: the next phase. SB Publishing (2018)
14. Leno, V., Dumas, M., La Rosa, M., Maggi, F.M., Polyvyanyy, A.: Automated discovery of data transformations for robotic process automation. arXiv preprint arXiv:2001.01007 (2020)
15. Leopold, H., van der Aa, H., Offenberg, J., Reijers, H.A.: Using hidden markov models for the accurate linguistic analysis of process model activity labels. Information Systems **83**, 30–39 (2019)
16. Leopold, H., van der Aa, H., Pittke, F., Raffel, M., Mendling, J., Reijers, H.A.: Searching textual and model-based process descriptions based on a unified data format. Software & Systems Modeling **18**(2), 1179–1194 (2019)
17. Leopold, H., van der Aa, H., Reijers, H.A.: Identifying candidate tasks for robotic process automation in textual process descriptions. In: Enterprise, business-process and information systems modeling, pp. 67–81. Springer (2018)
18. Leopold, H., Pittke, F., Mendling, J.: Automatic service derivation from business process model repositories via semantic technology. Journal of Systems and Software **108**, 134–147 (2015)
19. Leopold, H., Smirnov, S., Mendling, J.: On the refactoring of activity labels in business process models. Information Systems **37**(5), 443–459 (2012)
20. Schuler, K.K.: Verbnet: a broad-coverage, comprehensive verb lexicon. Ph.D. thesis, Philadelphia, PA, USA (2005)

21. Syed, R., Suriadi, S., Adams, M., Bandara, W., Leemans, S.J., Ouyang, C., ter Hofstede, A.H., van de Weerd, I., Wynn, M.T., Reijers, H.A.: Robotic process automation: Contemporary themes and challenges. Computers in Industry **115**, 103,162 (2020)
22. Wanner, J., Hofmann, A., Fischer, M., Imgrund, F., Janiesch, C., Geyer-Klingeberg, J.: Process selection in rpa projects–towards a quantifiable method of decision making (2019)
23. Wellmann, C., Stierle, M., Dunzer, S., Matzner, M.: A framework to evaluate the viability of robotic process automation for business process activities. In: RPA Forum (2020)